

Target Discrimination for SAR ATR Based on Scattering Center Feature and K-center One-Class Classification

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Abstract—Target discrimination plays an important role in the typical synthetic aperture radar (SAR) automatic target recognition (ATR) system. The conventional discrimination features roughly describing the difference between the target and the clutter are useful to discriminate the target from natural clutter, but these features may not well discriminate the target from the artificial clutter. To solve the above problem, a new algorithm for target discrimination is proposed based on the scattering center feature and the K-center one-class classification in this paper. The amplitudes and locations of the scattering centers have not been used as discrimination feature to our best knowledge, even though the distributions of the scattering centers associated with the targets are different from those of the natural clutter and artificial clutter. Then, since the scattering center feature is a typical kind of point set feature, we improve the K-center one-class classification based on Hausdorff distance (HD), where HD is a distance measure for the point set feature. Experimental results based on the measured MiniSAR dataset show that the discrimination performance of the scattering center feature is better than that of the conventional discrimination features, especially for the artificial clutter.

Index Terms—Synthetic aperture radar (SAR), target discrimination, scattering center, orthogonal matched pursuit (OMP), Hausdorff distance (HD), K-center one-class classification.

I. INTRODUCTION

SAR is widely used in military and civil fields, for its all-time operation, reduced sensitivity to weather conditions, and penetration capability through obstacles [1]. Generally, SAR usually generates large amounts of data in the form of two-dimensional imageries. Due to the large scale of these imageries and the relatively small size of targets within them, detecting and recognizing the targets via SAR ATR system becomes a serious issue [2].

A typical SAR ATR system is divided into three stages: detection, discrimination, and classification [3]–[5]. The first stage searches the entire image fast to find the regions of interest (ROI) that may contain potential targets. The cars are the target of interest in this paper. In this stage, targets as well as unavoidable clutter, both natural clutter and artificial

clutter, are detected. Here, the natural clutter refers to the clutter caused by natural things, like tree, glass, and so on. The artificial clutter refers to the clutter reflected by man-made structures or objects, such as buildings, helicopters, and so on. The second stage focus on removing the false alarms caused by natural clutter and artificial clutter, and maintaining the real targets. The third stage is to further reduce clutter false alarms and classify the target of interest. For the reason that the performance of the discrimination stage directly influences the whole SAR ATR system, this paper focuses on the discrimination stage.

Feature extraction plays an important role during target discrimination, thus, lots of discrimination features had been put forward, since the earliest three textural discriminate features [6] were proposed in 1989. These features describe the differences between targets and clutter from different properties, such as texture [6], size [3], [5], [7], contrast [5], [7], etc. But these features only roughly and partially describe the characteristic of the target, and they can't make full use of the abundant information contained in the SAR imageries. But these features only roughly describe the characteristic of the target, and may not well discriminate the target from the artificial clutter such as a building. For example, the fractal dimension feature provides a measure of the spatial dimensionality of the potential target [6]. This feature provides an estimate of the Hausdorff dimension of the spatial distribution of the top N brightest scatter points [5], and uses less information of the location of the brightest scatter points. Thus, such a simple feature cannot well discriminate the target from the artificial clutter, since both the target and the artificial clutter contain lots of brightest scatter points caused by metal.

To solve the above-mentioned issue, the scattering center feature is proposed to discriminate the target from the clutter, especially the artificial clutter.

According to the electromagnetic theory, high frequency response of an object can be approximated by a sum of responses from individual scattering centers [8]. Consequently, the scattering centers provide a physically relevant description of the target. In this paper, we use the attributes of these scattering centers, i.e. the amplitudes and locations, as the discrimination feature for the first time, called scattering center feature. This feature provides not only the amplitude but also location information of the scattering centers, and the locations of the scattering centers reflect the topology of the

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target in detail which is significant for target discrimination in SAR images. Moreover, the locations of the scattering centers associated with the targets are different from those of the natural clutter and artificial clutter.

The scattering center feature is firstly extracted from the sample by solving a sparse representation problem with the noise level as constraint. Then, since the directions and the positions of the targets maybe different, and the SAR data collected under different conditions will have different amplitude-scales, therefore, a preprocessing to rotate and shift the scattering centers is necessary before the scattering center feature is sent to a discriminator. Finally, the improved K-center one-class classification is developed as the discriminator to effectively discriminate the target from the clutter, especially the artificial clutter. Note that the distance measurement employed in traditional K-center one-class classification is the Euclidean distance measurement, which cannot measure the distance between two typical point sets. Thus, the Hausdorff distance (HD) is imported to improve the K-center one-class classification.

In this paper, a new algorithm for target discrimination is proposed, and this paper is distinguished by three key contributions. First, we attach importance to not only the natural clutter but also the artificial clutter to be discriminated from the targets in this discrimination algorithm, and effectively discriminating the artificial clutter from the targets is a difficult problem. Second, the scattering center is the first time being used as the discrimination feature, although it is useful for target discrimination. Third, it's a problem to find an effective and robust similarity measurement to evaluate the similarity of two scattering center features, since the scattering center feature is a typical point set. We use the Hausdorff distance as the distance measurement to solve the above problem. Experimental results show that compared with the conventional discrimination features, the proposed scattering center feature is more effective to discriminative the clutter from target, especially for the artificial clutter.

The remainder of the paper is organized as follows. We introduce the scattering center model and the method to extract the scattering center feature in Section II. Section III describes the well-known HD and the improved K-center one-class classification. The detailed experimental results based on the measured MiniSAR data are provided in Section IV, followed by conclusions in Section V.

II. FEATURE EXTRACTION

In this paper, the amplitudes and locations of the scattering centers are used as the discrimination feature to discriminate targets from clutter in SAR imagery. The scattering centers model is first described in the following. Then, the extracting method of the scattering centers is described in Section II-B. Finally, the preprocessing of the scattering center is introduced in Section II-C.

A. Scattering Center Model

At high frequency, according to the geometric theory of diffraction (GTD), the total back scattering response of an object

is well approximated as a sum of responses from individual scattering centers, or scatterers [8], [9]. The backscattered field of an individual scattering center is described as a function of frequency f and aspect ϕ angle, and the total backscattered field from a target is then modeled as the sum of these individual scattering centers.

$$E(f, \phi; \hat{\theta}) = \sum_{i=1}^N E_i(f, \phi; \theta_i) \quad (1)$$

$$E_i(f, \phi; \theta_i) = A_i \exp[-j \frac{4\pi f}{c} (x_i \cos \phi + y_i \sin \phi)] \quad (2)$$

where $E_i(f, \phi; \theta_i)$ is the response from the i th scattering center described by a parameter vector $\theta_i = [A_i, x_i, y_i]^T$, the parameters x_i and y_i are the downrange and crossrange locations, A_i is the scattering center amplitude. Then, c is the propagation velocity, N is the total number of individual scattering centers, $\hat{\theta} = [\theta_1, \dots, \theta_i, \dots, \theta_N]$ represents the parameter matrix of these N individual scattering centers, $()^T$ denotes the transpose operator.

B. Feature Extracting via Orthogonal Matching Pursuit Algorithm

As discussed in [10] for CS-based ISAR imaging and shown in Eq.(1), the signal can be approximated by the echoes from a few individual strong scattering centers, and thoes from the weak ones are regarded as noise. Considering this, we can extract scattering centers based on sparse representation.

$$\mathbf{s} = \mathbf{D}(\theta)\boldsymbol{\sigma} + \boldsymbol{\eta} \quad (3)$$

where $\mathbf{s}$ is the vectorization of measurements $E(f, \phi; \hat{\theta})$, $\boldsymbol{\sigma}$ is the coefficient vector, $\mathbf{D}(\theta)$ is a parameterized redundant dictionary, and each column of dictionary $\mathbf{D}(\theta)$ is the vectorization of measurements corresponding to one element in parameter set θ , and $\boldsymbol{\eta}$ represents the additive noise.

According to sparse representation theory, scattering centers can be extracted by solving the following sparse optimization problem:

$$\hat{\boldsymbol{\sigma}} = \arg \min_{\boldsymbol{\sigma}} \|\boldsymbol{\sigma}\|_0, \quad \text{s.t. } \|\mathbf{s} - \mathbf{D}(\theta)\boldsymbol{\sigma}\|_2^2 \leq \varepsilon \quad (4)$$

where $\|\cdot\|_0$ denotes l_0 norm and $\|\boldsymbol{\sigma}\|_0$ represents the number of non-zero entries in vector $\boldsymbol{\sigma}$, $\hat{\boldsymbol{\sigma}}$ is the estimated coefficient, $\varepsilon = \|\boldsymbol{\eta}\|_2^2$ is the noise level that can be estimated via some preprocessing methods [10]. In Eq.(4), the l_0 norm imposes that most elements of $\hat{\boldsymbol{\sigma}}$ are zero, in accordance with the sparsity of measurements $\mathbf{s}$ in the model parameter domain. The l_2 norm inequality constraint preserves the data fidelity of the solution with noise suppression under the noise level ε . The optimization problem (4) is nondeterministic polynomial time hard and difficult to solve [11], while its approximate solution can be efficiently obtained by greedy algorithms, such as basis pursuit [12], gradient pursuit [13], matching pursuit [14] and orthogonal matched pursuit (OMP) [15].

Here OMP algorithm is utilized to solve this minimum l_0 norm optimization problem. OMP is an iterative greedy algorithm that selects the column of $\mathbf{D}(\theta)$ which is the most correlated with current residual at each step. This column

is then added into the set of already selected columns. The residual is updated by subtracting the projection of $\mathbf{s}$ onto the linear subspace spanned by the already selected columns from the measurements $\mathbf{s}$, this operation guarantees the residual is always orthogonal to the already selected columns. The algorithm iterates until to the energy of the residual less than the noise level. The OMP algorithm for the scattering centers extraction is summarized in Algorithm 1.

In real application, the noise level could be estimated in advance. In SAR imagery, the estimation of the noise level is available since noise usually distributes evenly and there are many cells containing noise only [10]. If enough noise samples are given, the noise level can be estimated. Hence, the problem to estimate noise level is then transformed to discriminate noise cells from signal cells, which could be treated as a problem of target detection. There are many approaches presented to solve this problem, and we use method described in [16] to solve this problem. After noise cells are discriminated, noise level can be calculated by multiplying the averaging energy of these noise cells by the number of the cells.

Algorithm 1: OMP for the Scattering Centers Extraction

Input: The measurements $\mathbf{s}$, the estimated noise level ε , and the redundant parameterized dictionary $\mathbf{D}(\theta)$.

Output: The estimated parameters set $\hat{\theta}$

Initialization: The initial parameters set of the scattering centers $\hat{\theta} = \emptyset$, the residual signal $\mathbf{r} = \mathbf{s}$, and the iteration counter $t = 1$.

- 1: **while** $\|\mathbf{r}\|_2^2 > \varepsilon$ **do**
 - 2: Calculate correlation: $\mathbf{C}(\theta) = \mathbf{D}^H(\theta) \times \mathbf{r}$, where $(\cdot)^H$ denotes conjugate transpose.
 - 3: Estimate parameters: $\hat{\theta}_t = \arg \max_{\theta} (\mathbf{C}(\theta))$, $\hat{\theta} = \hat{\theta} \cup \hat{\theta}_t$.
 - 4: Estimate amplitudes: $\hat{\sigma} = \mathbf{D}^\dagger(\hat{\theta}) \times \mathbf{s}$, where $(\cdot)^\dagger$ denotes the Moore-Penrose pseudo-inverse, $\mathbf{D}(\hat{\theta})$ is the dictionary constructed by estimated parameters set $\hat{\theta}$.
 - 5: Update residual: $\mathbf{r} = \mathbf{s} - \mathbf{D}(\hat{\theta}) \times \hat{\sigma}$.
 - 6: $t = t + 1$
-

C. The Preprocessing of the Scattering Centers

As explained in Section I, the locations of the scattering centers cannot be used directly without any rotate and shift preprocessing, because of the direction and the position of the targets may be different. Thus, we employ the principal component analysis (PCA) to carry out the rotate process [17], [18], according to the first principal component, and shift the scatterers to the center of the chip.

Let $\mathbf{L} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n & \cdots & x_N \\ y_1 & y_2 & \cdots & y_n & \cdots & y_N \end{bmatrix}$ denotes the location matrix of the scattering centers extracted from a candidate target chip. In general, the mean of each row of $\mathbf{L}$ will be shifted to zero before PCA. The principle component can be obtained by decomposing the covariance matrix $\mathbf{C} = \mathbf{L}\mathbf{L}^T$. Here we employ the eigenvalue decomposition, i.e. $\mathbf{C} = \mathbf{E}\mathbf{\Lambda}\mathbf{E}^T$, where $\mathbf{\Lambda}$ is a diagonal matrix comprised by the eigenvalues, $\mathbf{E}$ is a matrix containing the corresponding

eigenvectors. The first principal component is the eigenvector corresponding to the largest eigenvalue, given as $\mathbf{p} = [a, b]^T$. Then, the angle between the first principal component and the vertical axis is $\varphi = \arctan \frac{a}{b}$, and then the rotate matrix is $\mathbf{G} = \begin{bmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{bmatrix}$. Note that, the mean of $\mathbf{L}$ has been shifted to zero. In this paper, the mean of $\mathbf{L}$ will be shifted to the center of the chip after the rotate process.

Furthermore, the SAR data collected under different conditions will have different amplitude-scales. To deal with this issue, the amplitudes of the scattering centers are normalized by dividing their l_2 -norm. Note that, the amplitudes of the scattering centers mentioned blow refer to the normalized amplitudes, and the locations of the scattering centers refer to the locations preprocessed.

III. AN IMPROVED K-CENTER ONE-CLASS CLASSIFICATION BASED ON HAUSDORFF DISTANCE

In this section, we first briefly review the Hausdorff distance [19], and the modified Hausdorff distance (MHD) [20]. Then, we improve the K-center one-class classification based on the MHD to make use of the typical point feature.

A. The Hausdorff Distance

The Hausdorff distance is a well-known method to measure the distance between two point sets. Given two finite point sets, $\mathbf{A} = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_P\}$ and $\mathbf{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_Q\}$, the Hausdorff distance is defined as

$$HD(\mathbf{A}, \mathbf{B}) = \max(h(\mathbf{A}, \mathbf{B}), h(\mathbf{B}, \mathbf{A})) \quad (5)$$

$$h(\mathbf{A}, \mathbf{B}) = \max_{\mathbf{a}_p \in \mathbf{A}} \min_{\mathbf{b}_q \in \mathbf{B}} md(\mathbf{a}_p, \mathbf{b}_q) \quad (6)$$

where the $h(\mathbf{A}, \mathbf{B})$ is called the directed Hausdorff distance from $\mathbf{A}$ to $\mathbf{B}$, $md(\mathbf{a}_p, \mathbf{b}_q)$ denotes the Mahalanobis distance between two points $\mathbf{a}_p$ and $\mathbf{b}_q$, the detailed method to calculate the Mahalanobis distance will be described in the following.

Considering the situation that $\mathbf{A}$ and $\mathbf{B}$ is quite similar, except a single point of $\mathbf{A}$ is far from every point of $\mathbf{B}$ caused by noise, then $h(\mathbf{A}, \mathbf{B})$ and $HD(\mathbf{A}, \mathbf{B})$ will be quite large. The sensitivity to an outlier or occlusion degrades the effectiveness of Hausdorff distance in practical tasks. Thus, some modified Hausdorff distances have been proposed, such as the modified Hausdorff distance(MHD) [20], the partial Hausdorff distance [21], and the weighted Hausdorff distance(WHD) [22]. Compared with other modified Hausdorff distances, the MHD are more suitable for object matching [20]. Thus, we employ the MHD to measure the distance between two point sets, and the directed distance of the MHD is defined as follows:

$$h_{\text{MHD}}(\mathbf{A}, \mathbf{B}) = \frac{1}{P} \sum_{p=1}^P \min_{\mathbf{b}_q \in \mathbf{B}} md(\mathbf{a}_p, \mathbf{b}_q) \quad (7)$$

The MHD is then defined as:

$$\text{MHD}(\mathbf{A}, \mathbf{B}) = \max(h_{\text{MHD}}(\mathbf{A}, \mathbf{B}), h_{\text{MHD}}(\mathbf{B}, \mathbf{A})) \quad (8)$$

As mentioned before, $md(\cdot, \cdot)$ denotes the Mahalanobis distance

$$md(\mathbf{a}_p, \mathbf{b}_q) = \sqrt{(\mathbf{a}_p - \mathbf{b}_q)^T \mathbf{\Sigma}^{-1} (\mathbf{a}_p - \mathbf{b}_q)} \quad (9)$$

where Σ generally is the measurement error covariance matrix. We calculate only the main diagonal for simplicity, thus Σ becomes a diagonal matrix, its diagonal elements are the measurement error variances of each feature in the feature vectors.

B. The Improved K-center One-Class Classification

As discussed in Section I, the improved K-center one-class classification is employed as the discriminator. The basic task of the K-center one-class classification is to define a boundary around the target class, such that it accepts as much of the target chips as possible, while it minimize the possibility of accepting clutter chips [23]. The boundary is described by two elements: the centers and the maximum cluster radius, note that the centers are subset of the target class.

In this paper, we propose the scattering center feature, which is a typical kind of point set feature, and the number of scattering centers extracted from chips maybe different. But the Euclidean distance employed in the traditional K-center one-class classification cannot measure the distance between two point sets with different number of points. Thus we improve the K-center one-class classification based on the MHD, and the improved K-center one-class classification is described in detail as follows.

Let $X = \{\theta_1, \theta_2, \dots, \theta_i, \dots, \theta_I\}$ denotes the training samples, and according to $p = \arg \min_k \text{MHD}(\theta_i, \mu_k)$, θ_i is categorized into cluster C_p , thus X is partitioned into K clusters $C = \{C_1, C_2, \dots, C_k, \dots, C_K\}$, and $J = \{\mu_1, \mu_2, \dots, \mu_k, \dots, \mu_K\}$ are the correspond K cluster centers, specially $J \subseteq X$. Where $\text{MHD}(\theta_i, \mu_k)$ is the modified Hausdorff distance between θ_1 and μ_k defined in Eq.8. To tightly describe the aforementioned boundary, the maximum distance of all minimum distances between training objects and the centers should be minimized. In other words, the following distance should be minimized:

$$d_r = \max_i (\min_k (\text{MHD}(\theta_i, \mu_k)^2)) \quad (10)$$

The K-center one-class classification uses a forward search strategy to train the centers [23]. In this strategy, different trails are repeated with different random choices for the initial centers, then each sample represents its cluster center, if the radius of the cluster can be decreased, exchange the former cluster center and the sample, until the centers can't be changed any more [24]. The radius of the cluster is defined as:

$$r(C_k) = \max_{\theta_i \in C_k} (\text{MHD}(\theta_i, \mu_k)) \quad (11)$$

When the centers have been trained, the maximum cluster radius is calculated by

$$r = \max_k r(C_k) \quad (12)$$

The K-center one-class classification focuses on the worst case samples (i.e. the samples with the largest distance to their corresponding cluster centers) and tries to optimize the centers

and the maximum cluster radius to accept all data [23]. Thus making the K-center one-class classification is sensitive to the worst case in the training samples, caused by measurement error or outliers. In order to deal with this situation, a reject ratio θ_r of the training samples should be given by the user. Hence the threshold is obtained by the following formulation:

$$\text{Thr} = (1 - \theta_r) \times r \quad (13)$$

During the testing stage, the distance from a testing sample $\hat{\theta}$ to the trained centers is defined as:

$$d(\hat{\theta}) = \min_k \text{MHD}(\hat{\theta}, \mu_k) \quad (14)$$

And the testing sample $\hat{\theta}$ is categorized into the target class if $d(\hat{\theta}) \leq \text{Thr}$.

IV. EXPERIMENTAL RESULTS

A. The Experimental SAR Image Dataset

In this section, we utilize the measured data to verify the proposed target discrimination method. The measured data we used here is the nine high-resolution Ku-band SAR images from the MiniSAR data set,¹ acquired by Sandia National Laboratories of America in 2005. The resolution of these images is 4 inch, and the size of these images is generally about 1638×2510, a little different between images. Two images from the nine images are shown in Fig.1. In these SAR images set, the cars are the targets of interest, and others, like buildings, trees, and grasslands are treated as clutter.

The candidate target chips are obtained from the nine high-resolution MiniSAR images via the most widely accepted two-parameter CFAR (constant false alarm rate) detector. The two-parameter CFAR detector is a pixel-by-pixel adaptive target detecting algorithm based on the assumption of Gaussian background. For every pixel in the image, the algorithm computed the mean and standard deviation surrounding the pixel. A guard region around the central pixel is used in this calculation to avoid the influence of target pixels. If the amplitude of the central pixel is above a CFAR threshold, the pixel is said to be due to a target. The CFAR threshold is determined by the mean and standard deviation of the surrounding region and the false alarm rate given by the user. If the pixels which pass this CFAR test are spatially close, they are clustered and said to be due to one target. In this paper, a 128×128 pixel region around the centroid of each such pixel cluster is then extracted. The detailed numbers of the extracted chips are shown in Table I, and examples of the candidate target chips are shown in the first row of Fig.2.

We select the candidate target chips with real targets to form the target samples, and the rest form the clutter samples. As shown in Fig.2, the candidate target chips are rather complex. First, for the cars are near to each other, the target samples may contain multiple car targets or some incomplete car targets. Second, different kinds of clutter are included in the clutter samples, such as buildings, trees, grasslands,

¹This data are downloaded from the webpage of the Sandia National Laboratories, and the website is <http://www.sandia.gov/radar/imagery>.

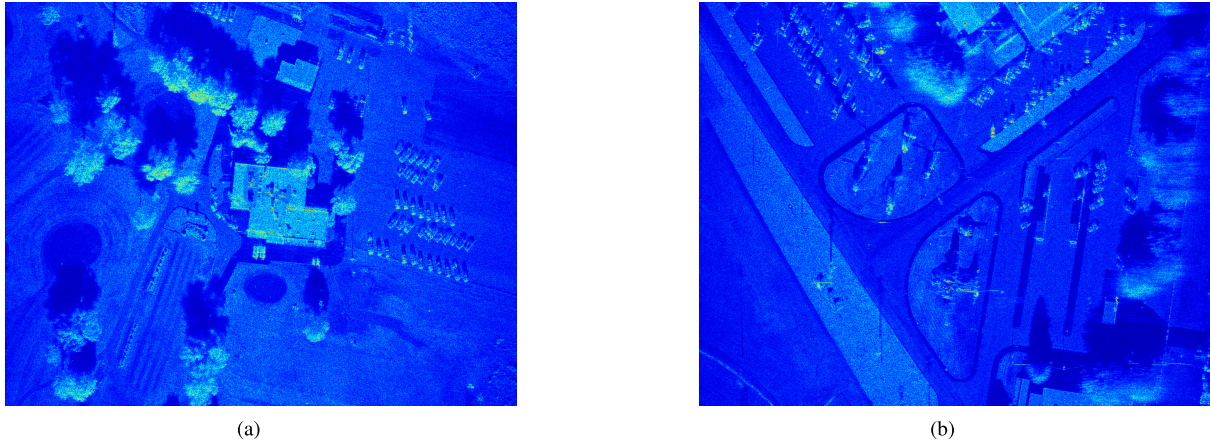


Fig. 1. Two examples of the nine high-resolution MiniSAR images

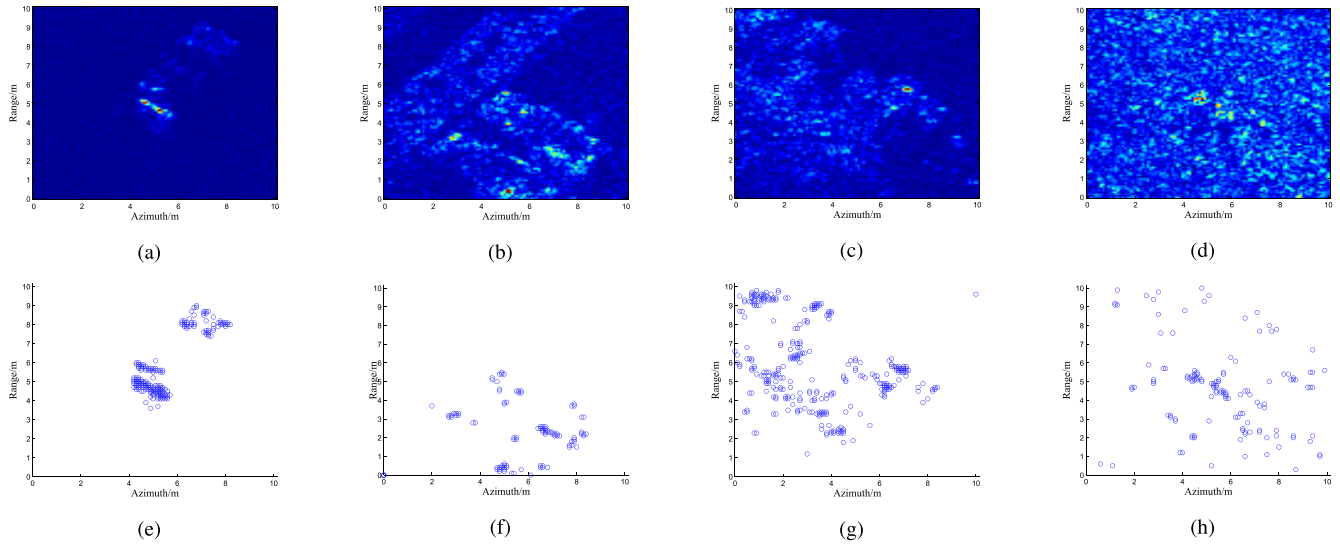


Fig. 2. Examples of the candidate target chips and the distribution of the scatterers, the first row is the candidate target chips and the second row is the distribution of the scatterers extracted from the corresponding chip in the first row. (a) is the target chip containing a single car, (b) is the target chip containing double cars, (c) is the natural clutter chip containing the tree crown, (d) is the natural clutter chip contain the grass land, (e)-(h) are the distribution of the scatterers extracted from (a)-(d) respectively.

helicopters, roads, and power poles. These two factors increase the difficulty of the target discrimination.

To fully evaluate the discrimination performance of the scattering center feature and compare the scattering center feature with other discrimination features, we divide the clutter samples into natural clutter and artificial clutter. The natural clutter chips refer to the chips containing tree, grass or other natural things. The artificial clutter chips refer to the chips containing building, power poles or other man-made structures or objects.

In the following, we adopt four conventional discrimination feature sets, including the old Lincoln features [5], [6], the new Lincoln features [25], the Bhanu's features [26], and the Gao's features [27], to compare with the scattering center feature to validate the effectiveness.

B. Analysis on Feature Separability

The scattering center feature is first extracted from training samples, examples of the distribution of the scatterers extracted are shown in the second row of the Fig.2. From the

second row of the Fig.2, we can see that the distribution of the scatterers is different between the target chips and the clutter chips.

Next, we utilize the ratio of the within-class distance to the between-class distance (RWTB) [28] to measure the linear separability of the proposed scattering center feature and other four conventional discrimination feature sets. The smaller within-class distance means the samples are close to each other in the category, and the larger between-class means that the two classes are far away from each other. Thus, an ideal feature will achieve the smaller RWTB value.

As explained in Section III-B, the Euclidean distance is not suitable to measure the distance between two point sets with different number of points. Thus we modify the traditional RWTB based on the MHD.

The within-class distance is defined as:

$$S_W = \frac{1}{N_1 + N_2} \sum_{k=1}^2 (N_k \times SW_k) \quad (15)$$

TABLE I
NUMBER OF DIFFERENT KINDS OF CHIPS EXTRACTED FROM NINE IMAGES

	Image 1	Image 2	Image 3	Image 4	Image 5	Image 6	Image 7	Image 8	Image 9	Total
Target	31	73	57	53	46	19	26	46	50	401
Natural clutter	37	27	16	57	114	64	65	23	30	433
Artificial clutter	58	107	93	7	26	57	27	12	21	408

TABLE II
RWTB VALUE OF DIFFERENT FEATURE SETS

Types of target and clutter	Old Lincoln features	New Lincoln features	Gao's features	Bhanu's features	Scattering Center Feature
Target VS Clutter	1.03	1.71	1.68	1.86	0.83
Target VS Natural clutter	1.65	5.33	2.50	4.81	1.32
Target VS Artificial clutter	1.80	7.89	8.01	6.43	1.46

where $SW_k = \frac{1}{N_k} \sum_{\mathbf{x} \in C'_k} \text{MHD}(\mathbf{x}, \mathbf{m}_k)$, $k = 1, 2$, $\mathbf{x}$ denotes the scattering center feature belonging to the k th class C'_k , N_k is the number of the samples in the k th class C'_k , $\mathbf{m}_k$ is the center of the k th class, i.e., $\mathbf{m}_k = \arg \min_{\mathbf{m}' \in C'_k} (\max_{\mathbf{x} \in C'_k} \text{MHD}(\mathbf{x}, \mathbf{m}'))$, and $\text{MHD}(\cdot, \cdot)$ is the modified Hausdorff distance denotes in Eq.8. The smaller within-class distance means the samples are close to each other in the category.

The between-class distance is defined as:

$$S_B = \frac{1}{N_1 + N_2} \sum_{k=1}^2 N_k \times \text{MHD}(\mathbf{m}_k, \mathbf{m}) \quad (16)$$

where $\mathbf{m}$ is the center of the two class C'_1 and C'_2 , i.e., $\mathbf{m} = \arg \min_{\mathbf{m}' \in (C'_1 \cup C'_2)} (\max_{\mathbf{x} \in (C'_1 \cup C'_2)} \text{MHD}(\mathbf{x}, \mathbf{m}'))$. The larger between-class means that the two classes are far away from each other.

Then, the RWTB is defined as:

$$J = S_w / S_B \quad (17)$$

Table II gives the RWTB value of the scattering center feature and the conventional discrimination feature sets. For example, the first row in Table II calculate the RWTB value of features between the target chips and clutter chips, this value measures the linear separability of the features to separate the target chips and clutter chips. The smaller RWTB value means the better sparability of the feature.

Based on the results in Table II, we can see that the scattering center feature is more discriminative than the conventional discrimination feature sets in the feature domain. What's more, the conventional discrimination feature sets merely suit the discrimination of the natural clutter. But the scattering center feature suits not only the discrimination of the natural clutter, but also the discrimination of the artificial clutter.

C. Parameter Setup

The improved K-center one-class discriminator introduced in section III-B includes two parameters that can be changed: (i) The number of clusters K , (ii) The reject ratio θ_r . In this section, we present our experimental results about the

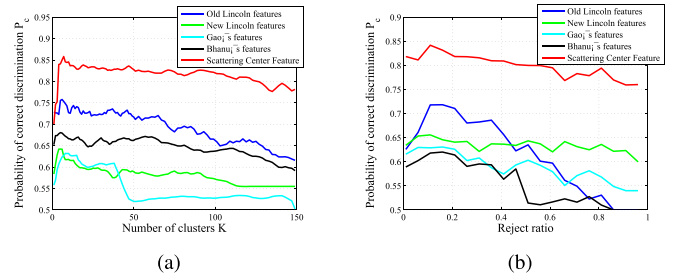


Fig. 3. The target discrimination performance versus the number of clusters K and the reject ratio θ_r .

impact of these parameters to performance using 5 fold cross-validation, the probability of correct discrimination p_c (defined in the following) is used to measure the performance of the discriminator on the test dataset.

In these parameter setup experiments, we random choose five SAR images from the nine MiniSAR images, the corresponding target samples form the training samples, the rest target samples from the other four images and all the clutter samples form the testing samples.

The effect of the number of clusters K and the reject ratio θ_r are given in Fig.3. When evaluate the effect of the number of clusters K (or the reject ratio θ_r), the reject ratio θ_r (or the number of clusters K) is fixed, and the value of the reject ratio θ_r (or the number of clusters K) is obtained by cross-validation.

As shown in Fig.3, the performance of the scattering center features and the conventional discrimination feature sets both achieved higher performance in a special interval. In Fig.3(a), since the larger the number of clusters K , the greater is the chance of overfitting the model, and the smaller the number of clusters K , the chance of underfitting the model is greater. Both overfitting and underfitting the model affect the performance of the K-center one-class discriminator. In Fig.3(b), the smaller the reject ratio θ_r , the more the clutter samples will be error divided as target, and the greater the reject ratio θ_r , the more the target samples will be wrongly classified as clutter. Both of the abovementioned condition reduced the performance.

TABLE III
DISCRIMINATION PERFORMANCE

Discrimination Performance	Old Lincoln features	New Lincoln features	Gao's features	Bhanu's features	Scattering Center Feature
pd(%)	75.42%±8.42%	76.50%±5.79%	46.15%±11.35%	74.70%±5.84%	83.21%±5.96%
pfa(%)	41.51%±5.66%	65.31%±8.08%	40.12%±6.25%	54.67%±4.32%	19.44%±6.23%
pc(%)	66.96%±3.47%	55.59%±4.52%	53.02%±3.79%	60.02%±2.94%	81.89%±3.11%

TABLE IV
DISCRIMINATION ERROR OF DIFFERENT TYPES OF CLUTTER

Types of clutter	Old Lincoln features	New Lincoln features	Gao's features	Bhanu's features	Scattering Center Feature
Natural clutter	20.97%±0.31%	57.95%±1.51%	38.34%±0.44%	49.46%±0.33%	11.02%±0.19%
Artificial clutter	54.49%±0.42%	73.12%±0.24%	42%±0.45%	60.19%±0.18%	33.16%±0.73%

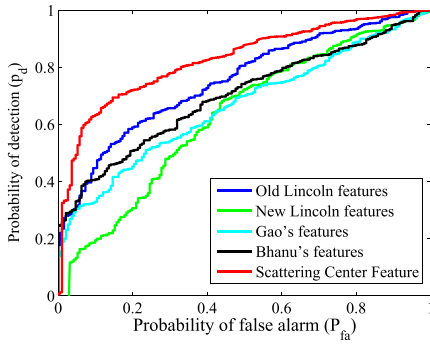


Fig. 4. The ROC curve

D. Discrimination Results and Analysis

The discrimination performance of the scattering center feature is evaluated in this part. For comparison, the performance of the conventional discrimination feature sets are also given. In order to evaluate the influence of the different partition of the training samples and testing samples on the discrimination performance, we repeat the discrimination experiment 126(C_9^5) times with different partition of the MiniSAR images. The discriminate results are shown in Table III, the receiver operator characteristic (ROC) curve of one discriminate experiment is shown in Fig.4, and the discriminate results of different types of clutter are given in Table IV.

Table III shows the discrimination performance in terms of the probability of detection p_d , probability of false alarm p_{fa} and probability of correct discrimination p_c . The probability of detection is defined as

$$p_d = \frac{N_d}{N_{Td}} \quad (18)$$

where N_d is the number of targets that are correctly discriminated, N_{Td} is the total number of targets.

The probability of false alarm p_{fa} is defined as

$$p_{fa} = \frac{N_{fa}}{N_{Tfa}} \quad (19)$$

where N_{fa} is the number of clutter false alarms that are categorized as targets. N_{Tfa} is the total number of clutter false alarms.

For the number of targets and the number of clutter used in testing samples are different, we defined the probability of correct discrimination p_c as

$$p_c = \frac{p_d + (1 - p_{fa})}{2} \quad (20)$$

As shown in Table III and Fig.4, the discriminate results of the proposed scattering center feature are superior to the four conventional discrimination feature sets, and the p_{fa} of the proposed feature is much smaller than the four conventional discrimination feature sets, especially the p_{fa} caused by artificial clutter shown in Table IV. The results also indicate that the proposed scattering center feature is suitable to discriminate not only natural clutter, but also artificial clutter from target.

E. Algorithm Analysis

1) *Robustness Analysis*: In order to analyze the robustness of the proposed discrimination algorithm, the performance of the scattering center feature and other conventional features are shown in Fig.5. Fig.5 shows the probability of correct discrimination p_c versus the average signal-to-clutter ratio(SCR). The SCR and average SCR are defined as

$$SCR = 10 \times \log_{10} \left(\frac{\sum P_s}{\sum P_c} \right) \quad (21)$$

$$avSCR = \frac{\sum_{i=1}^{N'} SCR_i}{N'} \quad (22)$$

where P_s denotes the power of the signal pixel divided by CFAR, P_c denotes the power of the clutter pixel, N' is the total number of chips.

As shown in Fig.5, when the $avSCR = 2\text{dB}$, the performance of the all features sharply decreases, but when the $avSCR \geq 4\text{dB}$, the influence of clutter is not so significant to the scattering center feature and old Lincoln feature. The performance of the scattering center feature is always better than other feature sets.

The scattering center feature is extracted by OMP with the noise level as constraint, thus the influence of clutter is little to the performance of the scattering center feature.

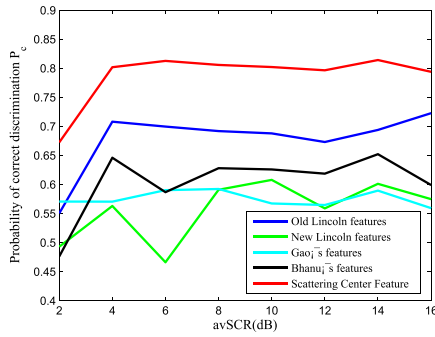


Fig. 5. Variation of the probability of correct discrimination with avSCR

TABLE V
AVERAGE TEST TIME OF DIFFERENT FEATURES

	Test time(s)
Old Lincoln features	1.57
New Lincoln features	1.62
Gao's features	1.35
Bhanu's features	1.78
Scattering Center Feature	512.3

2) *Algorithm Complexity Analysis*: In the test stage, the main processes are the feature extraction and the discrimination. The scattering center feature is extracted from a test chip by OMP, and the computational complexity of OMP is $o(M^3N^3P)$. Here, $M \times N$ represents the size of the test chip, P is the number of the proposed scattering centers. In this paper, $M = N = 128$. As shown in Eq.14, the test chip should calculate the MHD between the K clusters, and the computational complexity of this process is $o(KPQ)$. Here, Q is the average scattering center number of the K clusters. Thus, the main computational complexity of the proposed discrimination algorithm is $o(M^3N^3P) + o(KPQ)$.

3) *Time Consuming Analysis*: The average test time of the scattering center feature and other conventional features are shown in Table V. The experiments of all features are conducted using a PC with Intel(R) Core i5-3570 CPU of 3.4 GHz and memory of 4 GB. The program codes are written in MATLAB. The average test time of the scattering center feature is larger than other feature sets in the discrimination stage, but fewer false alarms can pass the proposed discrimination algorithm to the most time consuming stage, i.e. the classification stage.

Since OMP is used to extract the scattering center feature, the computation burden of our method is larger than that of other features. This is a disadvantage of our work. In the future work, we will try to use other method to lower the computational complexity and time consuming.

V. CONCLUSION

In this paper, we propose a novel target discrimination algorithm based on the scattering center feature and the improved K-center one-class classification. Feature extraction plays an important role in target discrimination, and the scattering center feature reveals the characteristic of the target

and clutter from the aspect of the electromagnetic scattering. To make full use of the typical point set feature and effectively discriminate target from the clutter, especially the artificial clutter, an improved K-center one-class classification based on the Hausdorff distance is introduced. Experiment results based on the MiniSAR data set show that the discrimination performance of the scattering center feature is more better than the conventional features, and the scattering center feature together with the improved K-center one-class classifier reaches a promising discrimination accuracy. Nevertheless, features more effectively reveal the electromagnetic scattering characteristic of the target should be investigated in the feature work.

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